Wiring Sheet – which cable goes where

Left / right / front / back are seen from BEHIND the robot, looking toward the intake. FRONT = intake + flower arm. BACK = launcher + camera.

Simple rule: everything on the robot's LEFT goes to the **CONTROL HUB**, everything on the RIGHT goes to the **EXPANSION HUB**. The launcher (middle) goes to the Control Hub.

CONTROL HUB (left side)

Port	Device	Encoder cable	Config type	Name (exact)	Done
Motor 0	Front-left drive wheel	Encoder 0	goBILDA 5202/3/4	`left_front_drive`	☐
Motor 1	Back-left drive wheel	Encoder 1	goBILDA 5202/3/4	`left_back_drive`	☐
Motor 2	Launcher / flywheel motor (middle, 1:1)	Encoder 2	goBILDA 5202/3/4	`launcher`	☐
Motor 3	SPARE – leave empty	–	–	`-`	☐
Servo 0	Left corner intake wheel servo	–	Continuous Rotation Servo	`left_intake_servo`	☐
USB 2.0	Logitech C270 webcam	–	Webcam	`Webcam 1`	☐
(built in)	IMU inside the Control Hub – nothing to plug	–	REV internal IMU	`imu`	☐
I2C 1	goBILDA Pinpoint (OPTIONAL, AUTO only) – pods into its X / Y ports	–	goBILDA Pinpoint Odometry Computer	`pinpoint`	☐

EXPANSION HUB (right side)

Port	Device	Encoder cable	Config type	Name (exact)	Done
Motor 0	Front-right drive wheel	Encoder 0	goBILDA 5202/3/4	`right_front_drive`	☐
Motor 1	Back-right drive wheel	Encoder 1	goBILDA 5202/3/4	`right_back_drive`	☐
Motor 2	Intake roller motor	NOT needed	goBILDA 5202/3/4	`intake`	☐
Motor 3	Windmill / indexer motor (435 RPM)	Encoder 3	goBILDA 5202/3/4	`indexer`	☐
Servo 0	Right corner intake wheel servo	–	Continuous Rotation Servo	`right_intake_servo`	☐

Port	Device	Encoder cable	Config type	Name (exact)	Done
Servo 1	Flower arm servo (SERVO mode, not continuous)	–	Servo	`flower_arm`	☐

Power and hub link

- Battery → main power switch → Control Hub **XT30 power** input.
- Control Hub's second XT30 → Expansion Hub XT30 (the Expansion Hub has no battery of its own).
- Control Hub **RS485** → Expansion Hub RS485 (either RS485 port on each hub).

Rules (check every cable)

- **Encoder number = motor number.** Motor in port 2 → its encoder cable in encoder port 2 on the SAME hub. Swapped encoder cables make the robot drive crazy.
- Encoders needed: all 4 drive motors, the launcher and the windmill/indexer. The intake motor needs NO encoder cable.
- **Servo plugs:** black (ground) wire on the pin marked – / ground on the hub. A plug turned around does nothing.
- The flower arm servo must be in SERVO (position) mode (goBILDA servo programmer) before it is installed.
- Webcam goes in the Control Hub's USB 2.0 port (the black one), not the blue USB 3.0 port.
- Keep cables away from the wheels, the intake rollers, the flywheel, the windmill and the flower arm's swing. Zip-tie them; leave a little slack at every moving part.
- Power the robot OFF before plugging or unplugging anything.

When you are done

- Driver Hub: robot configuration → check every name above is typed EXACTLY (lower case, underscores). A typo stops the program at INIT.
- Robot on a box, wheels free: run **BioBuzz Wiring Check**. It must say WIRING OK; every FAIL line says which cable to fix.
- Then run **BioBuzz System Check**.

Cable tags – cut out and tape/zip-tie one to each end of the cable

CONTROL HUB **Motor 0** left_front_drive	CONTROL HUB **Motor 1** left_back_drive	CONTROL HUB **Motor 2** launcher	CONTROL HUB **Servo 0** left_intake_servo	CONTROL HUB **USB 2.0** Webcam 1
CONTROL HUB **I2C 1** pinpoint	EXPANSION HUB **Motor 0** right_front_drive	EXPANSION HUB **Motor 1** right_back_drive	EXPANSION HUB **Motor 2** intake	EXPANSION HUB **Motor 3** indexer
EXPANSION HUB **Servo 0** right_intake_servo	EXPANSION HUB **Servo 1** flower_arm	CONTROL HUB **Encoder 0** left_front_drive (encoder)	CONTROL HUB **Encoder 1** left_back_drive (encoder)	CONTROL HUB **Encoder 2** launcher (encoder)
EXPANSION HUB **Encoder 0** right_front_drive (encoder)	EXPANSION HUB **Encoder 1** right_back_drive (encoder)	EXPANSION HUB **Encoder 3** indexer (encoder)	BOTH HUBS **RS485** Control Hub ↔ Expansion Hub	BOTH HUBS **XT30** Expansion Hub power (from Control Hub)